

# Discrete-time Martingales

## 1.1 Introduction

**Definition 1.1.** A discrete-time stochastic process  $M_n$  is a **martingale** if:

$$\mathbb{E}(M_n | \mathcal{F}_{n-1}) = M_{n-1},$$

where  $\mathcal{F}_n$  is a filtration including  $\sigma(M_k : k \leq n)$ , the *history of  $M_n$  at time  $n$* . ■

**Remark 1.1.** Unless otherwise stated, the martingale is supposed to satisfy the initial condition  $M_0 = 0$ . These processes are described as *mean-zero martingales*. ♦

**Example 1.1.** The martingale property is equivalent to:

$$\mathbb{E}(M_n | \mathcal{F}_m) = M_m \text{ for } m < n.$$

To see this, suppose  $m \leq n - 1$ , then by iterated expectation:

$$\mathbb{E}(M_n | \mathcal{F}_m) \stackrel{*}{=} \mathbb{E}\{\mathbb{E}(M_n | \mathcal{F}_{n-1}) | \mathcal{F}_m\} = \mathbb{E}(M_{n-1} | \mathcal{F}_m).$$

Note that  $\stackrel{*}{=}$  holds since  $\mathcal{F}_m \subseteq \mathcal{F}_{n-1}$ . If  $m = n - 1$ , then the result has been shown. Otherwise, continue in like manner until:

$$\mathbb{E}(M_n | \mathcal{F}_m) = \cdots = \mathbb{E}(M_{m+1} | \mathcal{F}_m) = M_m. \spadesuit$$

**Proposition 1.1.** Martingales have constant mean  $\mathbb{E}(M_n) = \mathbb{E}(M_0)$ . ♦

**Proof.**

$$\mathbb{E}(M_n) = \mathbb{E}\{\mathbb{E}(M_n | \mathcal{F}_0)\} = \mathbb{E}(M_0). \blacksquare$$

**Proposition 1.2.** Martingales have *uncorrelated increments*:

$$\mathbb{C}(M_m, M_n - M_m) = 0 \quad \diamond$$

**Proof.**

$$\mathbb{C}(M_m, M_n - M_m) = \mathbb{E}\{M_m(M_n - M_m)\} - \mathbb{E}(M_m)\mathbb{E}(M_n - M_m).$$

The first term vanishes since:

$$\begin{aligned}\mathbb{E}\{M_m(M_n - M_m)\} &= \mathbb{E}[\mathbb{E}\{M_m(M_n - M_m)|\mathcal{F}_m\}] \\ &= \mathbb{E}[M_m\{\mathbb{E}(M_n|\mathcal{F}_m) - M_m\}] \\ &= \mathbb{E}\{M_m(M_m - M_m)\} = 0.\end{aligned}$$

The second term vanishes since:

$$\begin{aligned}\mathbb{E}(M_n - M_m) &= \mathbb{E}\{\mathbb{E}(M_n - M_m|\mathcal{F}_m)\} \\ &= \mathbb{E}\{\mathbb{E}(M_n|\mathcal{F}_m) - M_m\} \\ &= \mathbb{E}(M_m - M_m) = 0.\end{aligned}$$

■

**Example 1.2.** Suppose  $Z_i$  are independent and identically distributed (IID) random variables with  $\mathbb{E}(Z_i) = 0$  and  $\mathbb{V}(Z_i) = \sigma^2$ . Define  $Y_n$  as the partial sum process  $\sum_{i=1}^n Z_i$ , and let  $\mathcal{F}_n = \sigma(Y_k : k \leq n)$ .  $Y_n$  is a mean-zero martingale. To see this, write:

$$Y_n = \sum_{i=1}^n Z_i = Z_n + \sum_{i=1}^{n-1} Z_i = Z_n + Y_{n-1}.$$

Taking the expectation conditional on the history:

$$\mathbb{E}(Y_n|\mathcal{F}_{n-1}) = \mathbb{E}(Z_n|\mathcal{F}_{n-1}) + \mathbb{E}(Y_{n-1}|\mathcal{F}_{n-1}) \stackrel{*}{=} \mathbb{E}(Z_n) + Y_{n-1} = Y_{n-1},$$

where  $\stackrel{*}{=}$  holds because the  $Z_n$  is independent of  $\mathcal{F}_{n-1}$ . Finally, the marginal expectation of  $Y_n$  is:

$$\mathbb{E}(Y_n) = \sum_{i=1}^n \mathbb{E}(Z_i) = 0.$$

♠

## 1.2 Martingale Difference Sequence

**Definition 1.2.** A discrete-time stochastic process  $\Delta M_n$  is a **martingale difference sequence** if:

$$\mathbb{E}(\Delta M_n|\mathcal{F}_{n-1}) = 0.$$

■

**Proposition 1.3.** If  $M_n$  is a martingale, then  $\Delta M_n = M_n - M_{n-1}$  is a martingale difference sequence. ♦

**Proof.**

$$\begin{aligned}\mathbb{E}(\Delta M_n | \mathcal{F}_{n-1}) &= \mathbb{E}(M_n - M_{n-1} | \mathcal{F}_{n-1}) \\ &= \mathbb{E}(M_n | \mathcal{F}_{n-1}) - M_{n-1} \\ &= M_{n-1} - M_{n-1} = 0.\end{aligned}$$

■

### 1.3 Variation Processes

**Definition 1.3.** Suppose  $M_n$  is a discrete-time mean-zero martingale. The **predictable variation** of  $M_n$  is the sum of the conditional variances of the increments:

$$\langle M \rangle_n = \sum_{m=1}^n \mathbb{V}(\Delta M_m | \mathcal{F}_{m-1}) = \sum_{m=1}^n \mathbb{E}\{(M_m - M_{m-1})^2 | \mathcal{F}_{m-1}\}.$$

The **optional variation** of  $M_n$  is the sum of the squared martingale differences:

$$[M]_n = \sum_{m=1}^n (\Delta M_m)^2 = \sum_{m=1}^n (M_m - M_{m-1})^2.$$

The predictable and optional variations are initialized at zero:

$$\langle M \rangle_0 = 0, \quad [M]_0 = 0.$$

■

**Proposition 1.4.** Suppose  $M_n$  is a discrete-time mean-zero martingale. Then, the *compensated process*  $M_n^2 - \langle M \rangle_n$  is also mean-zero martingale. ◆

**Proof.** Write the square  $M_n^2$  as :

$$M_n^2 = (M_n - M_{n-1} + M_{n-1})^2 = (\Delta M_n)^2 + M_{n-1}^2 + 2(\Delta M_n)M_{n-1},$$

and the predictable variation as:

$$\langle M \rangle_n = \mathbb{E}\{(M_n - M_{n-1})^2 | \mathcal{F}_{n-1}\} + \langle M \rangle_{n-1} = \mathbb{E}\{(\Delta M_n)^2 | \mathcal{F}_{n-1}\} + \langle M \rangle_{n-1}.$$

The expectation of the difference satisfies:

$$\begin{aligned}\mathbb{E}(M_n^2 - \langle M \rangle_n | \mathcal{F}_{n-1}) &= \mathbb{E}\{(\Delta M_n)^2 + M_{n-1}^2 + 2(\Delta M_n)M_{n-1} - (\Delta M_n)^2 - \langle M \rangle_{n-1} | \mathcal{F}_{n-1}\} \\ &= \mathbb{E}(M_{n-1}^2 - \langle M \rangle_{n-1} | \mathcal{F}_{n-1}) + 2\mathbb{E}\{(\Delta M_n)M_{n-1} | \mathcal{F}_{n-1}\} \\ &= M_{n-1}^2 - \langle M \rangle_{n-1} + 2\mathbb{E}(\Delta M_n | \mathcal{F}_{n-1})M_{n-1} = M_{n-1}^2 - \langle M \rangle_{n-1}.\end{aligned}$$

Finally,  $M_n^2 - \langle M \rangle_n$  is mean-zero because:

$$\mathbb{E}\{M_n^2 - \langle M \rangle_n\} = \mathbb{E}\{M_0^2 - \langle M \rangle_0\} = \mathbb{E}(0) = 0.$$

■

**Problem 1.1.** Suppose  $M_n$  is a discrete-time mean-zero martingale. Prove that the compensated process  $M_n^2 - [M]_n$  is also a mean-zero martingale. ♠

**Proposition 1.5.** Suppose  $M_n$  is a discrete-time mean-zero martingale, then:

$$\mathbb{V}(M_n) = \mathbb{E}\langle M \rangle_n = \mathbb{E}[M]_n.$$

◆

**Proof.** Since  $\mathbb{E}(M_n) = 0$ , the variance  $\mathbb{V}(M_n) = \mathbb{E}(M_n^2)$ . Now, since the compensated process  $M_n^2 - \langle M \rangle_n$  is a mean-zero martingale:

$$\mathbb{E}\{M_n^2 - \langle M \rangle_n\} = 0 \implies \mathbb{E}(M_n^2) = \mathbb{E}\langle M \rangle_n.$$

Likewise for  $M_n^2 - [M]_n$ . ■

## 1.4 Covariation Processes

**Definition 1.4.** Suppose  $M_{1n}$  and  $M_{2n}$  are discrete-time mean-zero martingales. The **predictable covariation** of  $M_{1n}$  and  $M_{2n}$  is the sum of the conditional covariance of the increments:

$$\langle M_1, M_2 \rangle_n = \sum_{m=1}^n \mathbb{C}(\Delta M_{1m} \Delta M_{2m} | \mathcal{F}_{m-1}) = \sum_{m=1}^n \mathbb{E}(\Delta M_{1m} \Delta M_{2m} | \mathcal{F}_{m-1}).$$

The **optional covariation** of  $M_{1n}$  and  $M_{2n}$  is the sum of the joint increments:

$$[M_1, M_2]_n = \sum_{m=1}^n \Delta M_{1m} \Delta M_{2m}.$$

■

**Proposition 1.6.** Suppose  $M_{1n}$  and  $M_{2n}$  are discrete-time mean-zero martingales. Then, the compensated process  $M_{1n}M_{2n} - \langle M_1, M_2 \rangle_n$  is also a mean-zero martingale. ◆

**Proof.** Write the product  $M_{1n}M_{2n}$  as:

$$\begin{aligned} M_{1n}M_{2n} &= (\Delta M_{1n} + M_{1(n-1)})(\Delta M_{2n} + M_{2(n-1)}) \\ &= \Delta M_{1n}\Delta M_{2n} + \Delta M_{1n}M_{2(n-1)} + M_{1(n-1)}\Delta M_{2n} + M_{1(n-1)}M_{2(n-1)}, \end{aligned}$$

and the predictable covariation  $\langle M_1, M_2 \rangle_n$  as:

$$\langle M_1, M_2 \rangle_n = \mathbb{E}(\Delta M_{1n}\Delta M_{2n} | \mathcal{F}_{n-1}) + \langle M_1, M_2 \rangle_{n-1}.$$

Now the expectation of the difference is expressible as:

$$\begin{aligned}
& \mathbb{E}(M_{1n}M_{2n} - \langle M_1, M_2 \rangle_n | \mathcal{F}_{n-1}) \\
&= \mathbb{E}\{ \Delta M_{1n} \Delta M_{2n} + \Delta M_{1n} M_{2(n-1)} + M_{1(n-1)} \Delta M_{2n} + M_{1(n-1)} M_{2(n-1)} \\
&\quad - \Delta M_{1n} \Delta M_{2n} - \langle M_1, M_2 \rangle_{n-1} | \mathcal{F}_{n-1} \} \\
&= M_{1(n-1)} M_{2(n-1)} - \langle M_1, M_2 \rangle_{n-1} \\
&\quad + \mathbb{E}(\Delta M_{1n} | \mathcal{F}_{n-1}) M_{2(n-1)} + M_{1(n-1)} \mathbb{E}(\Delta M_{2n} | \mathcal{F}_{n-1}) \\
&= M_{1(n-1)} M_{2(n-1)} - \langle M_1, M_2 \rangle_{n-1}.
\end{aligned}$$

Finally,  $M_{1n}M_{2n} - \langle M_1, M_2 \rangle_n$  is mean-zero because:

$$\mathbb{E}\{M_{1(n-1)}M_{2(n-1)} - \langle M_1, M_2 \rangle_{n-1}\} = \mathbb{E}\{M_{10}M_{20} - \langle M_1, M_2 \rangle_0\} = \mathbb{E}(0) = 0.$$

■

**Problem 1.2.** Suppose  $M_{1n}$  and  $M_{2n}$  are discrete-time mean-zero martingales. Prove that the compensated process  $M_{1n}M_{2n} - [M_1, M_2]_n$  is also a mean-zero martingale. ♠

**Proposition 1.7.** Suppose  $M_{1n}$  and  $M_{2n}$  are discrete-time mean-zero martingales, then:

$$\mathbb{C}(M_{1n}, M_{2n}) = \mathbb{E}\langle M_1, M_2 \rangle_n = \mathbb{E}[M_1, M_2]_n.$$

◆

**Proof.** Since  $\mathbb{E}(M_{1n}) = \mathbb{E}(M_{2n}) = 0$ , the covariance  $\mathbb{C}(M_{1n}, M_{2n}) = \mathbb{E}(M_{1n}M_{2n})$ . Now, since  $M_{1n}M_{2n} - \langle M_1, M_2 \rangle_n$  is a mean-zero martingale:

$$\mathbb{E}\{M_{1n}M_{2n} - \langle M_1, M_2 \rangle_n\} = 0 \implies \mathbb{E}(M_{1n}M_{2n}) = \mathbb{E}\langle M_1, M_2 \rangle_n.$$

Likewise for  $M_{1n}M_{2n} - [M_1, M_2]_n$ .

■

## 1.5 Transformations

**Definition 1.5.** A discrete-time process  $H_n$  is **predictable** if  $H_n$  is measurable with respect to  $\mathcal{F}_{n-1}$ . ■

**Definition 1.6.** The **transformation** of a process  $X_n$  by  $H_n$ , written  $Z = H \bullet X$ , is the process:

$$Z_n = \sum_{m=1}^n H_m \Delta X_m. \quad (1.1)$$

■

**Proposition 1.8 (Martingale Transformation).** The transformation of a discrete-time mean-zero martingale  $M_n$  by a predictable process remains a mean-zero martingale. ◆

**Proof.** We show that  $\Delta Z_n$  is a martingale difference sequence.

$$\begin{aligned}\mathbb{E}(\Delta Z_n | \mathcal{F}_{n-1}) &= \mathbb{E}(Z_n - Z_{n-1} | \mathcal{F}_{n-1}) \\ &= \mathbb{E}(H_n \Delta M_n | \mathcal{F}_{n-1}) \\ &\stackrel{i.}{=} H_n \mathbb{E}(\Delta M_n | \mathcal{F}_{n-1}) \stackrel{ii.}{=} 0,\end{aligned}$$

where  $\stackrel{i.}{=}$  holds because  $H_n$  is predictable, and  $\stackrel{ii.}{=}$  holds because  $\Delta M_n$  is a martingale difference sequence. ■

**Proposition 1.9.** The predictable variation of a transformation satisfies:

$$\langle H \bullet X \rangle = H^2 \bullet \langle X \rangle. \quad (1.2)$$
◆

**Proof.** First, observe that:

$$\begin{aligned}\Delta(H \bullet X)_n &= (H \bullet X)_n - (H \bullet X)_{n-1} \\ &= \sum_{m=1}^n H_m \Delta X_m - \sum_{m=1}^{n-1} H_m \Delta X_m = H_n \Delta X_n.\end{aligned}$$

Also:

$$\begin{aligned}\Delta \langle X \rangle_n &= \langle X \rangle_n - \langle X \rangle_{n-1} \\ &= \sum_{m=1}^n \mathbb{V}(\Delta M_m | \mathcal{F}_{m-1}) - \sum_{m=1}^{n-1} \mathbb{V}(\Delta M_m | \mathcal{F}_{m-1}) = \mathbb{V}(\Delta M_n | \mathcal{F}_{n-1}).\end{aligned}$$

Therefore:

$$\begin{aligned}\langle H \bullet X \rangle_n &= \sum_{m=1}^n \mathbb{V}\{\Delta(H \bullet X)_m | \mathcal{F}_{m-1}\} \\ &= \sum_{m=1}^n \mathbb{V}(H_m \Delta X_m | \mathcal{F}_{m-1}) \\ &= \sum_{m=1}^n H_m^2 \mathbb{V}(\Delta X_m | \mathcal{F}_{m-1}) = \sum_{m=1}^n H_m^2 \Delta \langle X \rangle_m.\end{aligned}$$
■

**Problem 1.3.** Verify that the optional variation of a transformation satisfies:

$$[H \bullet X] = H^2 \bullet [X].$$

♠

**Definition 1.7.** A random variable  $\tau$  is a **stopping time** if the event  $\{\tau = t\}$  is decidable (measurable) with respect to  $\mathcal{F}_t$ . ■

**Proposition 1.10.** If  $M_n$  is a mean-zero martingale and  $\tau$  is a stopping time, then the *stopped process*  $M_n^\tau = M_{n \wedge \tau}$  is a mean-zero martingale. ◆

**Proof.** Define the predictable process:

$$H_n = \begin{cases} 0, & \tau \leq n-1, \\ 1, & \tau > n-1 \end{cases} = \mathbb{I}(\tau > n-1).$$

The stopped martingale is expressible as:

$$M_{n \wedge \tau} = \sum_{m=1}^n H_m \Delta M_m = H \bullet M.$$

To see this, note that if  $\tau \geq n$

$$\sum_{m=1}^n \Delta M_m = (M_n - M_{n-1}) + (M_{n-1} - M_{n-2}) + \cdots + (M_2 - M_1) + M_1 = M_n.$$

If  $\tau < n$  was the stopping time, then:

$$\sum_{m=1}^n H_m \Delta M_m = 0 + \cdots + (M_\tau - M_{\tau-1}) + \cdots + (M_2 - M_1) + M_1 = M_\tau.$$

Since the stopped martingale  $M_{n \wedge \tau}$  is the transformation of a mean-zero martingale by a predictable process,  $M_{n \wedge \tau}$  is again a mean-zero martingale. ■

## Continuous-time Martingales

### 2.1 Introduction

**Definition 2.1.** A continuous-time stochastic process  $X(t)$  is **adapted** to a *filtration*  $\mathcal{F}(t)$  if for each  $t$ ,  $X(t)$  is measurable with respect to  $\mathcal{F}(t)$ . ■

**Definition 2.2.** A continuous-time stochastic process  $M_t$  is a **martingale** if:

$$\mathbb{E}\{M(t) | \mathcal{F}(s)\} = M(s) \text{ for } s \leq t,$$

where  $\mathcal{F}(t)$  is a filtration including  $\sigma\{M(s) : s \leq t\}$ . ■

**Remark 2.1.** Each realization of a continuous-time stochastic process is called at **sample path**. Unless otherwise stated, continuous-time martingales are assumed to have *càdlàg* sample paths, to satisfy the initial condition  $M(0) = 0$ , and to have a finite time horization  $\mathcal{T} = [0, \tau]$ . Many of the properties of discrete-time martingales carry over to continuous-time. ◆

**Discussion 2.1.** The martingale property may be expressed in differential form as:

$$\mathbb{E}\{dM(t)|\mathcal{F}(t-)\} = 0,$$

where  $dM(t)$  is the increment of the martingale over the time interval  $[t, t + dt)$  and  $\mathcal{F}(t-)$  is the history just before time  $t$ . Compare this with the discrete-time result:

$$\mathbb{E}(\Delta M_n | \mathcal{F}_{n-1}) = 0.$$



**Proposition 2.1.** As in discrete-time, continuous-time martingales have constant mean and *uncorrelated increments*:

$$\mathbb{C}\{M(t) - M(s), M(v) - M(u)\} = 0,$$

for  $s < t < u < v$ . ◆

**Proof.** To demonstrate constant mean:

$$\mathbb{E}\{M(t)\} = \mathbb{E}[\mathbb{E}\{M(t)|\mathcal{F}(0)\}] = \mathbb{E}\{M(0)\}.$$

For uncorrelated increments, first observe that:

$$\mathbb{E}\{M(t) - M(s)\} = \mathbb{E}\{M(v) - M(u)\} = 0$$

due to constant mean. Thus:

$$\mathbb{C}\{M(t) - M(s), M(v) - M(u)\} = \mathbb{E}[\{M(t) - M(s)\}\{M(v) - M(u)\}].$$

By iterated expectation:

$$\begin{aligned} \mathbb{E}[\{M(t) - M(s)\}\{M(v) - M(u)\}] &= \mathbb{E}(\mathbb{E}[\{M(t) - M(s)\}\{M(v) - M(u)\} | \mathcal{F}(u)]) \\ &= \mathbb{E}(\{M(t) - M(s)\}[\mathbb{E}\{M(v)|\mathcal{F}(u)\} - M(u)]) \\ &= \mathbb{E}[\{M(t) - M(s)\}\{M(u) - M(u)\}] = 0. \end{aligned}$$





## 2.2 Variation Processes

**Definition 2.3.** Suppose  $M(t)$  is a continuous-time mean-zero martingale. The **predictable variation** of  $M(t)$  is:

$$\langle M \rangle(t) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{V}\{\Delta M_k | \mathcal{F}(\delta_{k-1})\}, \quad (2.1)$$

where  $\delta_k = (k/n)t$ , and  $\Delta M_k = M(\delta_k) - M(\delta_{k-1})$ . Note that the  $\delta_k$  partition the interval  $[0, t]$  into  $n$  sub-intervals of length  $t/n$ .

The **optional variation** of  $M(t)$  is:

$$[M](t) = \lim_{n \rightarrow \infty} \sum_{k=1}^n (\Delta M_k)^2, \quad (2.2)$$

where  $\Delta M_k = M(\delta_k) - M(\delta_{k-1})$  as for the predictable variation. ■

**Discussion 2.2.** In differential form:

$$d\langle M \rangle(t) = \mathbb{V}\{dM(t) | \mathcal{F}(t-)\}.$$

Compare this with the discrete-time result, obtained in the proof of (1.2), that:

$$\Delta \langle M \rangle_n = \mathbb{V}(\Delta M_n | \mathcal{F}_{n-1}).$$



**Proposition 2.2.** Suppose  $M(t)$  is a continuous-time mean-zero martingale. Then, the following *compensated processes* are also mean-zero martingales:

$$M^2(t) - \langle M \rangle(t), \quad M^2(t) - [M](t).$$

Consequently,

$$\mathbb{V}\{M(t)\} = \mathbb{E}\{M^2(t)\} = \mathbb{E}\langle M \rangle(t) = \mathbb{E}[M](t).$$



## 2.3 Covariation Process

**Definition 2.4.** Suppose  $M_1(t)$  and  $M_2(t)$  are continuous-time mean-zero martingales. The **predictable covariation** of  $M_1$  and  $M_2$  is:

$$\langle M_1, M_2 \rangle(t) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{C}\{\Delta M_{1k} \Delta M_{2k} | \mathcal{F}(\delta_{k-1})\},$$

where  $\delta_k = (k/n)t$ , and  $\Delta_{jk} = M_j(\delta_k) - M_j(\delta_{k-1})$ . The **optional covariation** of  $M_1$  and  $M_2$  is:

$$[M_1, M_2](t) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \Delta M_{1k} \Delta M_{2k}.$$

■

**Proposition 2.3.** Suppose  $M_1(t)$  and  $M_2(t)$  are continuous-time mean-zero martingales. Then the following *compensated processes* are also mean-zero martingales:

$$M_1 M_2 - \langle M_1, M_2 \rangle, \quad M_1 M_2 - [M_1, M_2].$$

Consequently,

$$\mathbb{C}\{M_1(t), M_2(t)\} = \mathbb{E}\{M_1(t)M_2(t)\} = \mathbb{E}\langle M_1, M_2 \rangle(t) = \mathbb{E}[M_1, M_2](t).$$

◆

**Proposition 2.4.** Similar to the covariance operator for random variables, the predictable and optional covariation processes are bilinear:

$$\begin{aligned} \langle M_1 + M_2 \rangle &= \langle M_1 \rangle + \langle M_2 \rangle + 2\langle M_1, M_2 \rangle, \\ [M_1 + M_2] &= [M_1] + [M_2] + 2[M_1, M_2]. \end{aligned}$$

◆

## 2.4 Stochastic Integrals

**Definition 2.5.** A stochastic process  $H(t)$  is **predictable** if  $H(t)$  is adapted to the filtration  $\mathcal{F}(t)$  and  $H(t)$  has *left-continuous* sample paths. ■

**Definition 2.6.** The **stochastic integral** of a predictable process with respect to a martingale may be defined as:

$$I(t) = \int_0^t H(s) dM(s) = \lim_{n \rightarrow \infty} \sum_{k=1}^n H_k \Delta M_k,$$

where  $\delta_k = (k/n)t$  partitions the interval  $[0, t]$  into  $n$  sub-intervals of length  $t/n$ ,  $H_k$  is  $H(\delta_{k-1})$ , and  $\Delta M_k = M(\delta_k) - M(\delta_{k-1})$ . ■

**Theorem 2.1 (Martingale Transformation).** The stochastic integral  $I(t)$  of a predictable process  $H(t)$  with respect to a mean-zero martingale  $M(t)$  remains a mean-zero martingale. □

**Remark 2.2.** This is analogous to preservation of the martingale property under transformation  $(H \bullet M)$  of a discrete time martingale  $M_n$  by a predictable process  $H_n$ . ♦

**Proposition 2.5.** The predictable and optional covariations of the stochastic integral of a predictable process with respect to a mean-zero martingale obeys the following rules:

$$\begin{aligned} \left\langle \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right\rangle &= \int H_1 H_2(t) d\langle M_1, M_2 \rangle(t) \\ \left[ \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right] &= \int H_1 H_2(t) d[M_1, M_2](t). \end{aligned}$$

In the case that  $H_1 = H_2 = H$ :

$$\begin{aligned} \left\langle \int H(t) dM(t) \right\rangle &= \int H^2(t) d\langle M \rangle(t) \\ \left[ \int H(t) dM(t) \right] &= \int H^2(t) d[M](t) \end{aligned}$$

♦

## 2.5 Doob-Meyer Decomposition

**Definition 2.7.** A continuous-time stochastic process  $N(t)$  is a **sub-martingale** if it tends to increase as time passes:

$$\mathbb{E}\{N(t) | \mathcal{F}(s)\} \geq N(s) \text{ for } t > s.$$

■

**Theorem 2.2.** A continuous-time sub-martingale  $N(t)$  (e.g. a counting process) can be uniquely decomposed as:

$$N(t) = A(t) + M(t), \tag{2.3}$$

where  $A(t)$  is a non-decreasing predictable process (the *compensator*), and  $M(t)$  is a mean-zero martingale. □

**Discussion 2.3.** In differential form, the Doob-Meyer decomposition is expressible as:

$$dN(t) = dA(t) + dM(t).$$

Upon taking the conditional expectation:

$$\mathbb{E}\{dN(t) | \mathcal{F}(t-)\} = \mathbb{E}\{dA(t) | \mathcal{F}(t-)\} + \mathbb{E}\{dM(t) | \mathcal{F}(t-)\}.$$

Since  $A(t)$  is a predictable process and  $M(t)$  is a martingale:

$$\mathbb{E}\{dN(t)|\mathcal{F}(t-)\} = dA(t).$$

The increments of the martingale are thus:

$$\begin{aligned} dM(t) &= dN(t) - dA(t) \\ &= dN(t) - \mathbb{E}\{dN(t)|\mathcal{F}(t-)\}. \end{aligned}$$



## 2.6 Poisson Process

**Example 2.1.** Suppose  $N(t)$  is the number of events by time  $t$  in a standard Poisson process with rate  $\lambda$ . Define the compensated process:

$$M(t) = N(t) - \lambda t.$$

By the independent increments property:

$$\mathbb{E}\{M(t) - M(s)|\mathcal{F}(s)\} = \mathbb{E}\{M(t) - M(s)\} = \mathbb{E}\{N(t) - N(s)\} - \lambda(t - s) = 0.$$

Therefore:

$$0 = \mathbb{E}\{M(t) - M(s)|\mathcal{F}(s)\} = \mathbb{E}\{M(t)|\mathcal{F}(s)\} - M(s).$$

Combined with  $M(0) = 0$ , this shows  $M(t)$  is a mean-zero martingale. Now consider the square process:

$$M^2(t) = N^2(t) + (\lambda t)^2 - 2\lambda t N(t)$$

Consider the process  $M^2(t) - \lambda t$ :

$$\mathbb{E}\{M^2(t) - \lambda t|\mathcal{F}(s)\} = \mathbb{E}\{N^2(t)|\mathcal{F}(s)\} + (\lambda t)^2 - 2\lambda t \mathbb{E}\{N(t)|\mathcal{F}(s)\} - \lambda t.$$

The first expectation reduces as:

$$\begin{aligned} \mathbb{E}\{N^2(t)|\mathcal{F}(s)\} &= \mathbb{E}[\{N(t) - N(s) + N(s)\}^2|\mathcal{F}(s)] \\ &= \mathbb{E}[\{N(t) - N(s)\}^2|\mathcal{F}(s)] + N^2(s) + 2N(s)\mathbb{E}\{N(t) - N(s)|\mathcal{F}(s)\} \end{aligned}$$

By the independent increments property:

$$\begin{aligned} \mathbb{E}[\{N(t) - N(s)\}^2|\mathcal{F}(s)] &= \mathbb{E}[\{N(t) - N(s)\}^2] \\ &= \mathbb{V}\{N(t - s)\} + \mathbb{E}^2\{N(t - s)\} \\ &= \lambda(t - s) + \lambda^2(t - s)^2. \end{aligned}$$

The second expectation reduces to:

$$\begin{aligned}\mathbb{E}\{N(t)|\mathcal{F}(s)\} &= \mathbb{E}\{N(t) - N(s) + N(s)|\mathcal{F}(s)\} \\ &= \mathbb{E}\{N(t) - N(s)\} + N(s) \\ &= \lambda(t - s) + N(s)\end{aligned}$$

Overall:

$$\begin{aligned}\mathbb{E}\{M^2(t) - \lambda t|\mathcal{F}(s)\} &= \lambda(t - s) + \lambda^2(t - s)^2 + N^2(s) + 2\lambda(t - s)N(s) \\ &\quad + (\lambda t)^2 - 2\lambda t\{\lambda(t - s) + N(s)\} - \lambda t \\ &= N^2(s) + \lambda^2 s^2 - 2\lambda s N(s) - \lambda s \\ &= M^2(s) - \lambda s.\end{aligned}$$

Conclude that  $\langle M \rangle(t) = \lambda t$ , which gives:

$$\mathbb{E}\{M^2(t)\} = \mathbb{E}\langle M \rangle(t) = \lambda t.$$



## 2.7 Counting Process

**Definition 2.8.** A **counting process** is a continuous-time *càdlàg* process with increments of size 1 at event times. ■

**Discussion 2.4.** Since any counting process  $N(t)$  is a sub-martingale, by the Doob-Meyer decomposition, there exists a unique predictable process  $\Lambda(t)$ , the *cumulative intensity*, such that  $M(t) = N(t) - \Lambda(t)$  is a mean-zero martingale. If  $\Lambda(t)$  is absolutely continuous, there exists a predictable *intensity* process such that:

$$\Lambda(t) = \int_0^t \lambda(s) ds.$$

In differential form:

$$dM(t) = dN(t) - \lambda(t)dt.$$

Upon taking the conditional expectation:

$$\begin{aligned}\mathbb{E}\{dM(t)|\mathcal{F}(t-)\} &= \mathbb{E}\{dN(t)|\mathcal{F}(t-)\} - \mathbb{E}\{\lambda(t)dt|\mathcal{F}(t-)\} \\ 0 &= \mathbb{E}\{dN(t)|\mathcal{F}(t-)\} - \lambda(t)dt.\end{aligned}$$

Finally, since  $N(t)$  has unit increments,  $dN(t)$  is Bernoulli, and:

$$\lambda(t)dt = \mathbb{E}\{dN(t)|\mathcal{F}(t-)\} = \mathbb{P}\{dN(t) = 1|\mathcal{F}(t-)\}.$$

For the optional variation of a counting process martingale, note that as the partition of  $[0, t]$  becomes infinitely fine, only the jumps  $dN(t)$ , which are of size 1, will contribute to (2.2). This motivates:

$$[M](t) = N(t).$$

For the predictable variation,

$$\begin{aligned} d\langle M \rangle(t) &= \mathbb{V}\{dM(t) | \mathcal{F}(t-)\} \\ &= \mathbb{V}\{dN(t) - \lambda(t)dt | \mathcal{F}(t-)\} \\ &\stackrel{*}{=} \mathbb{V}\{dN(t) | \mathcal{F}(t-)\}. \end{aligned}$$

where  $\stackrel{*}{=}$  holds since  $\lambda(t)$  is predictable. Now, since  $dN(t)$  is Bernoulli:

$$d\langle M \rangle(t) = \lambda(t)dt\{1 - \lambda(t)dt\} = \lambda(t)dt + o(dt).$$

This motivates the relation:

$$\langle M \rangle(t) = \int_0^t \lambda(s)ds = \Lambda(t).$$

That is, the predictable variation of a counting process martingale  $M(t) = N(t) - \Lambda(t)$  is its cumulative intensity. ♠

**Theorem 2.3.** If  $N(t)$  is a counting process with cumulative intensity  $\Lambda(t)$ , then the following are mean-zero martingales:

$$\begin{aligned} M(t) &= N(t) - \Lambda(t), \\ M^2(t) &- \Lambda(t) \end{aligned}$$

□

**Definition 2.9.** Two martingales  $M_1$  and  $M_2$  are **orthogonal** if the predictable covariation is zero:

$$\langle M_1, M_2 \rangle(t) = 0.$$

Note that, for orthogonal martingales, the product  $M_1(t)M_2(t)$  is itself a mean-zero martingale. ■

**Proposition 2.6.** If  $N_1(t)$  and  $N_2(t)$  are counting processes that cannot jump simultaneously, then the corresponding martingales  $M_1(t)$  and  $M_2(t)$  are orthogonal. ♦

**Discussion 2.5.** Suppose  $N_1(t), \dots, N_n(t)$  are counting processes, no two of which can jump simultaneously, then the corresponding martingales  $M_1(t), \dots, M_n(t)$  are orthogonal. Since the predictable covariation obeys:

$$\left\langle \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right\rangle = \int H_1 H_2(t) d\langle M_1, M_2 \rangle(t),$$

the corresponding stochastic integrals  $\int H_1(t) dM_1(t), \dots, \int H_n(t) dM_n(t)$  are orthogonal. From bilinearity of the predictable variation:

$$\left\langle \sum_{i=1}^n \int_0^t H_i(s) dM_i(s) \right\rangle = \sum_{i=1}^n \int_0^t H_i^2(s) d\langle M_i \rangle(s) = \sum_{i=1}^n \int_0^t H_i^2(s) \lambda_i(s) ds.$$

For the optional variation:

$$\left[ \sum_{i=1}^n \int_0^t H_i(s) dM_i(s) \right] = \sum_{i=1}^n \int_0^t H_i^2(s) d[M_i](s) = \sum_{i=1}^n \int_0^t H_i^2(s) dN_i(s).$$

♠

## 2.8 Martingale Central Limit Theorem

**Definition 2.10.** The **Brownian motion** process  $W(t)$  has continuous sample paths and independent, stationary increments. The increment over the time interval  $(s, t]$  is normally distributed with:

$$\mathbb{E}\{W(t) - W(s)\} = 0, \quad \mathbb{V}\{W(t) - W(s)\} = t - s.$$

If  $\alpha(t)$  is a strictly increasing and continuous remapping of time, with  $\sigma^2(0) = 0$ , then  $U(t) = W\{\sigma^2(t)\}$  is again a Brownian motion. The increment over the time interval  $(s, t]$  is normally distributed with:

$$\mathbb{E}\{U(t) - U(s)\} = 0, \quad \mathbb{V}\{U(t) - U(s)\} = \sigma^2(t) - \sigma^2(s).$$

The process  $U(t) = W\{\sigma^2(t)\}$  is a *Gaussian martingale*. ■

**Proposition 2.7.** Let  $U(t) = W\{\sigma^2(t)\}$ , where  $\sigma^2(t)$  is a strictly increasing and continuous remapping of time, with  $\alpha(0) = 0$ . Then  $U(t)$  is a mean-zero martingale, with predictable variation  $\langle U \rangle(t) = \sigma^2(t)$ . ♦

**Proof.**

$$\begin{aligned} \mathbb{E}\{U(t) | \mathcal{F}(s)\} &= \mathbb{E}\{U(t) - U(s) + U(s) | \mathcal{F}(s)\} \\ &= \mathbb{E}\{U(t) - U(s) | \mathcal{F}(s)\} + U(s) \\ &= \mathbb{E}\{U(t) - U(s)\} + U(s) = U(s). \end{aligned}$$

Thus  $U(t)$  is a martingale. Note too that:

$$\mathbb{E}\{U(t) - U(s) | \mathcal{F}(s)\} = 0,$$

and that  $\sigma^2(t)$  is deterministic. Now consider the process  $U^2(t) - \sigma^2(t)$ :

$$\begin{aligned} \mathbb{E}\{U^2(t) - \sigma^2(t) | \mathcal{F}(s)\} &= \mathbb{E}[\{U(t) - U(s) + U(s)\}^2 | \mathcal{F}(s)] - \sigma^2(t) \\ &= \mathbb{E}[\{U(t) - U(s)\}^2 | \mathcal{F}(s)] \\ &\quad + U^2(s) + 2U(s)\mathbb{E}\{U(t) - U(s) | \mathcal{F}(s)\} - \sigma^2(t). \end{aligned}$$

From the above, the cross term vanishes. The second moment evaluates to:

$$\begin{aligned} \mathbb{E}[\{U(t) - U(s)\}^2 | \mathcal{F}(s)] &= \mathbb{E}[\{U(t) - U(s)\}^2] \\ &= \mathbb{V}\{U(t) - U(s)\} + \mathbb{E}\{U(t) - U(s)\}^2 \\ &= \sigma^2(t) - \sigma^2(s) + 0. \end{aligned}$$

Overall:

$$\mathbb{E}\{U^2(t) - \sigma^2(t) | \mathcal{F}(s)\} = \sigma^2(t) - \sigma^2(s) + U^2(s) - \sigma^2(t) = U^2(s) - \sigma^2(s).$$

Since  $\sigma^2(0) = 0$ ,  $U^2(t) - \sigma^2(t)$  is a mean-zero martingale, and:

$$\mathbb{V}\{U(t)\} = \mathbb{E}\{U^2(t)\} = \mathbb{E}\langle U \rangle(t) = \sigma^2(t).$$

■

**Theorem 2.4 (Martingale CLT).** Suppose  $M^{(n)}(t)$  is a sequence of mean-zero martingales defined on  $[0, \tau]$ , and for any  $\epsilon > 0$  let  $M_\epsilon^{(n)}(t)$  denote the martingale containing all jumps of  $M^{(n)}(t)$  that are of size greater than  $\epsilon$ . If the following conditions hold:

- i.  $\langle M^{(n)} \rangle \xrightarrow{p} \sigma^2(t)$  for all  $t \in [0, \tau]$  as  $n \rightarrow \infty$ , where  $\sigma^2(t)$  is a strictly increasing continuous function with  $\sigma^2(0) = 0$ .
- ii.  $\langle M_\epsilon^{(n)} \rangle \xrightarrow{p} 0$  for all  $t \in [0, \tau]$  for any  $\epsilon > 0$  as  $n \rightarrow \infty$ .

Then,  $M^{(n)}(t)$  converges weakly to a Gaussian martingale:

$$M^{(n)}(t) \rightsquigarrow W\{\sigma^2(t)\}.$$

□

**Example 2.2.** Suppose  $N_1^{(n)}(t), \dots, N_n^{(n)}(t)$  are counting processes, no two of which can jump simultaneously. Recall that the corresponding martingales  $M_1^{(n)}(t), \dots, M_n^{(n)}(t)$  are orthogonal. Define the process:

$$U^{(n)}(t) = \sum_{i=1}^n U_i^{(n)}(t) = \sum_{i=1}^n \int_0^t H_i^{(n)}(s) dM_i^{(n)}(s),$$



where  $H_i^{(n)}$  is a predictable process, and:

$$M_i^{(n)} = N_i^{(n)} - \int_0^t \lambda_i^{(n)}(s) ds$$

is a counting process martingale. The requirements for the martingale CLT are that:

$$\langle U^{(n)} \rangle(t) = \sum_{i=1}^n \int_0^t \{H_i^{(n)}(t)\}^2 d\lambda_i^{(n)}(s) \xrightarrow{p} \alpha(t) \text{ for } \forall t \in [0, \tau].$$

$$\langle U_\epsilon^{(n)} \rangle(t) = \sum_{i=1}^n \int_0^t \{H_i^{(n)}(t)\}^2 \mathbb{I}\{|H_i^{(n)}(s)| > \epsilon\} d\lambda_i^{(n)}(s) \xrightarrow{p} 0 \text{ for } \forall \epsilon > 0, \forall t \in [0, \tau].$$



**Example 2.3.** For  $n \rightarrow \infty$ , the path of  $U^{(n)}(t)$  on the interval  $[0, \tau]$  is approximated by that of a particle undergoing time-transformed Brownian motion  $U(t) = W\{\alpha(t)\}$ . For a continuous functional  $f$ , the behavior of  $f \circ W\{\alpha(t)\}$  approximates that of  $f \circ U^{(n)}(s)$ :

$$\{f \circ U^{(n)}(s) : s \in [0, \tau]\} \rightsquigarrow \{f \circ U(s) : s \in [0, \tau]\}.$$

For example, the supremum of  $|U^{(n)}(t)|$  may be approximated by that of  $|U(t)|$ :

$$\sup_{t \in [0, \tau]} |U^{(n)}(t)| \rightsquigarrow \sup_{t \in [0, \tau]} |U(t)|.$$



## Summary

- In discrete time, the **martingale property** is  $\mathbb{E}(M_n | \mathcal{F}_{n-1}) = M_{n-1}$ . In continuous time  $\mathbb{E}\{M(t) | \mathcal{F}(s)\} = M(s)$  for  $s \leq t$ .

- Martingales have constant mean  $\mathbb{E}(M_n) = M_0$  and *uncorrelated increments*:

$$\mathbb{C}\{M(t) - M(s), M(v) - M(u)\} = 0.$$

- In discrete-time, the predictable **covariation** of a mean-zero martingale  $M_n$  is defined as:

$$\langle M_1, M_2 \rangle_n = \sum_{m=1}^n \mathbb{C}(\Delta M_{1m} \Delta M_{2m} | \mathcal{F}_{m-1}) = \sum_{m=1}^n \mathbb{E}(\Delta M_{1m} \Delta M_{2m} | \mathcal{F}_{m-1}).$$

The optional **covariation** is defined as:

$$[M_1, M_2]_n = \sum_{m=1}^n \Delta M_{1m} \Delta M_{2m}.$$

The predictable and optional **variations** are obtained by taking  $M_1 = M_2 = M$ . The continuous-time covariations at time  $t$  are obtained by taking the limit as the partition of  $[0, t]$  becomes infinitely fine.

The covariance of a martingale is the expectation of its predictable (or optional) variation. In continuous time:

$$\mathbb{C}\{M_1(t), M_2(t)\} = \mathbb{E}\{M_1(t)M_2(t)\} = E\langle M_1, M_2 \rangle(t) = E[M_1, M_2](t).$$

Taking  $M_1 = M_2 = M$ :

$$\mathbb{V}\{M(t)\} = \mathbb{E}\{M^2(t)\} = E\langle M \rangle(t) = E[M](t).$$

- In discrete-time, the **transformation** of a mean-zero martingale by a predictable process is:

$$Z_n = \sum_{m=1}^n H_m \Delta X_m.$$

In continuous-time, by taking the limit as the partition of  $[0, t]$  becomes infinitely fine, the transformation becomes a **stochastic integral**:

$$I(t) = \int_0^t H(s) dM(s).$$

In either case, the transformation or stochastic integral of a mean-zero martingale by a predictable process remains a mean-zero martingale.

- The predictable covariation of stochastic integrals obeys:

$$\left\langle \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right\rangle = \int H_1 H_2(t) d\langle M_1, M_2 \rangle(t).$$

Similarly, for the optional variation:

$$\left[ \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right] = \int H_1 H_2(t) d[M_1, M_2](t)$$

- A continuous-time stochastic process  $N(t)$  is a **sub-martingale** if it tends to increase as time passes:

$$\mathbb{E}\{N(t) | \mathcal{F}(s)\} \geq N(s) \text{ for } t > s.$$

**(Doob-Meyer Decomposition)** A continuous-time sub-martingale  $N(t)$  can be uniquely decomposed as:

$$N(t) = A(t) + M(t), \tag{3.1}$$

where  $A(t)$  is a non-decreasing predictable process (the *compensator*), and  $M(t)$  is a mean-zero martingale.

- A **counting process** is a continuous-time *càdlàg* process with increments of size 1 at event times. By Doob-Meyer, if  $N(t)$  is a counting process, there exists a unique predictable process, termed the *cumulative intensity*, such that  $M(t) = N(t) - \Lambda(t)$  is a mean-zero martingale.

The optional variation of a counting process martingale is the counting process:

$$[M] = N(t).$$

The predictable variation is its cumulative intensity:

$$\langle M \rangle(t) = \Lambda(t).$$

- Counting processes  $N_1(t)$  and  $N_2(t)$  are **orthogonal** if the predictable covariation of the corresponding martingales is zero:

$$\langle M_1, M_2 \rangle(t) = 0.$$

If two counting processes cannot jump simultaneously, then they are orthogonal.

- The **Brownian motion** process  $W(t)$  has continuous sample paths, independent stationary increments, and the increment over the time interval  $(s, t]$  is normally distributed with:

$$\mathbb{E}\{W(t) - W(s)\} = 0, \quad \mathbb{V}\{W(t) - W(s)\} = t - s.$$

If  $\alpha(t)$  is a strictly increasing continuous remapping of time with  $\alpha(0) = 0$ , then  $U(t) = W\{\alpha(t)\}$  is again Brownian motion, with increments:

$$\mathbb{E}\{U(t) - U(s)\} = 0, \quad \mathbb{V}\{U(t) - U(s)\} = \alpha(t) - \alpha(s).$$

The process  $U(t) = W\{\alpha(t)\}$  is a *Gaussian martingale*.

- **(Martingale Central Limit Theorem)** Suppose  $M^{(n)}(t)$  is a sequence of mean-zero martingales defined on  $[0, \tau]$ , and for any  $\epsilon > 0$  let  $M_\epsilon^{(n)}(t)$  denote the martingale containing all jumps of  $M^{(n)}(t)$  that are of size greater than  $\epsilon$ . If the following conditions hold:

- $\langle M^{(n)} \rangle \xrightarrow{p} \sigma^2(t)$  for all  $t \in [0, \tau]$  as  $n \rightarrow \infty$ , where  $\sigma^2(t)$  is a strictly increasing continuous function with  $\sigma^2(0) = 0$ .
- $\langle M_\epsilon^{(n)} \rangle \xrightarrow{p} 0$  for all  $t \in [0, \tau]$  for any  $\epsilon > 0$  as  $n \rightarrow \infty$ .

Then,  $M^{(n)}(t)$  converges weakly to a Gaussian martingale:

$$M^{(n)}(t) \rightsquigarrow W\{\sigma^2(t)\}.$$